Caleb Elder

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Education

University of Georgia December 2026

3.98 GPA | B.S. Computer Science, Minor in Mathematics

Achievements: Hollingsworth Award in Linear Algebra, President's Honor Roll, Dean's List, Zell Miller Scholarship

Relevant Coursework: Algorithms, Data Structures, Graph Theory, Applied Linear Algebra, Discrete Mathematics, Systems Programming, Computer Architecture, Computer Networks, Theory of Computing, Mathematical Analysis, Software Engineering, Software Development

Technical Skills

Programming Languages (Proficient): Java, Python, C
Programming Languages (Familiar): TypeScript, JavaScript, SQL, C++, HTML/CSS, LaTeX
Human Languages: Proficient: Spanish | Intermediate: French, Chinese | Conversational: German, Russian, Korean
Cloud & Developer Tools: Azure (App Service, Functions, ACI), Git, GitHub Actions (CI/CD), Docker, Linux, Bash
Frameworks & Libraries: React, Node.js, Express, Supabase (Postgres RLS), Playwright, NumPy, Pandas, PyTorch
Data & Protocols: REST APIs, MySQL, PostgreSQL, JSON, Gson, MSAL/Azure AD Auth

Experience

Fetch Freight May 2026 – Present

Software Engineering Intern *Birmingham, AL*

- Shipped 5 production systems in 8 weeks (Azure, Node.js/TypeScript, React, PostgreSQL), including a full-stack negotiation platform with automated writes into the company's transportation management system
- Co-developed an AI document-processing pipeline (Claude vision) that classifies, matches, and autonomously files hundreds of carrier emails per day, with feedback loops that learn from human corrections
- Reverse-engineered undocumented legacy systems and rebuilt them as tested cloud services, proving penny-perfect parity across 291 orders and surfacing $720K+ in past-due receivables on the first production run
- Established the team's safe-rollout playbook (shadow modes, idempotent send ledgers, feature flags) and owned security fixes including a stored-XSS patch and catching a vendor link exposing edit access to company data

AI4STEM (VIPR Program) August 2025 – December 2025

Undergraduate Research *Athens, GA*

- Researched LLM automated assessment models, sourcing benchmark datasets and designing a multimodal rubric integrating text and image data
- Developed Python data pipelines to process spreadsheet metrics into structured JSON, hitting the OpenAI API to generate automated, reasoning-based grading scores
- Evaluated LLM system performance using Cohen's Kappa, achieving 0.66-0.69 (substantial agreement) across 3 trials correlating model outputs against human teacher scores; presented findings at the CURO Symposium

Projects

AI Language Learning Platform | *Next.js, TypeScript, Supabase, Vercel AI SDK* Summer 2026

- Built a full-stack Spanish/French learning platform (CEFR A1–C2) with AI stories, streaming tutoring, dictionary/conjugations, spaced-repetition vocab, and multiplayer games
- Implemented a streaming AI tutor (GPT-4o-mini) with abortable client streaming, session persistence, and background personalization that re-analyzes learner traits and pre-queues conversation starters
- Designed Supabase Postgres schema with row-level security and Realtime chat/pictionary rooms; added `after()` generation queues so stories and chat starters pre-generate without blocking responses
- Shipped structured AI pipelines with Zod schemas across story generation, grammar drills, pronunciation scoring (ElevenLabs + Web Speech), and an offline crossword seed generator with grid validation

Cinema E-Booking System | *React, TypeScript, Flask, MySQL, Docker* Spring 2026

- Collaborated on a full-stack cinema booking platform with movie search, interactive seat selection, checkout, payments, promotions, admin portals, and email confirmations
- Built a typed React/TypeScript frontend against a Flask REST API and MySQL schema, containerized with Docker Compose for consistent team development
- Applied design patterns (Facade, Builder, Adapter, Proxy) to keep booking, payment, and admin flows modular and maintainable across a multi-developer codebase